

量子抗性架構從晶片開始

Quantum-Resistant Architecture Begins at the Chip

Chris Lin | March 2026

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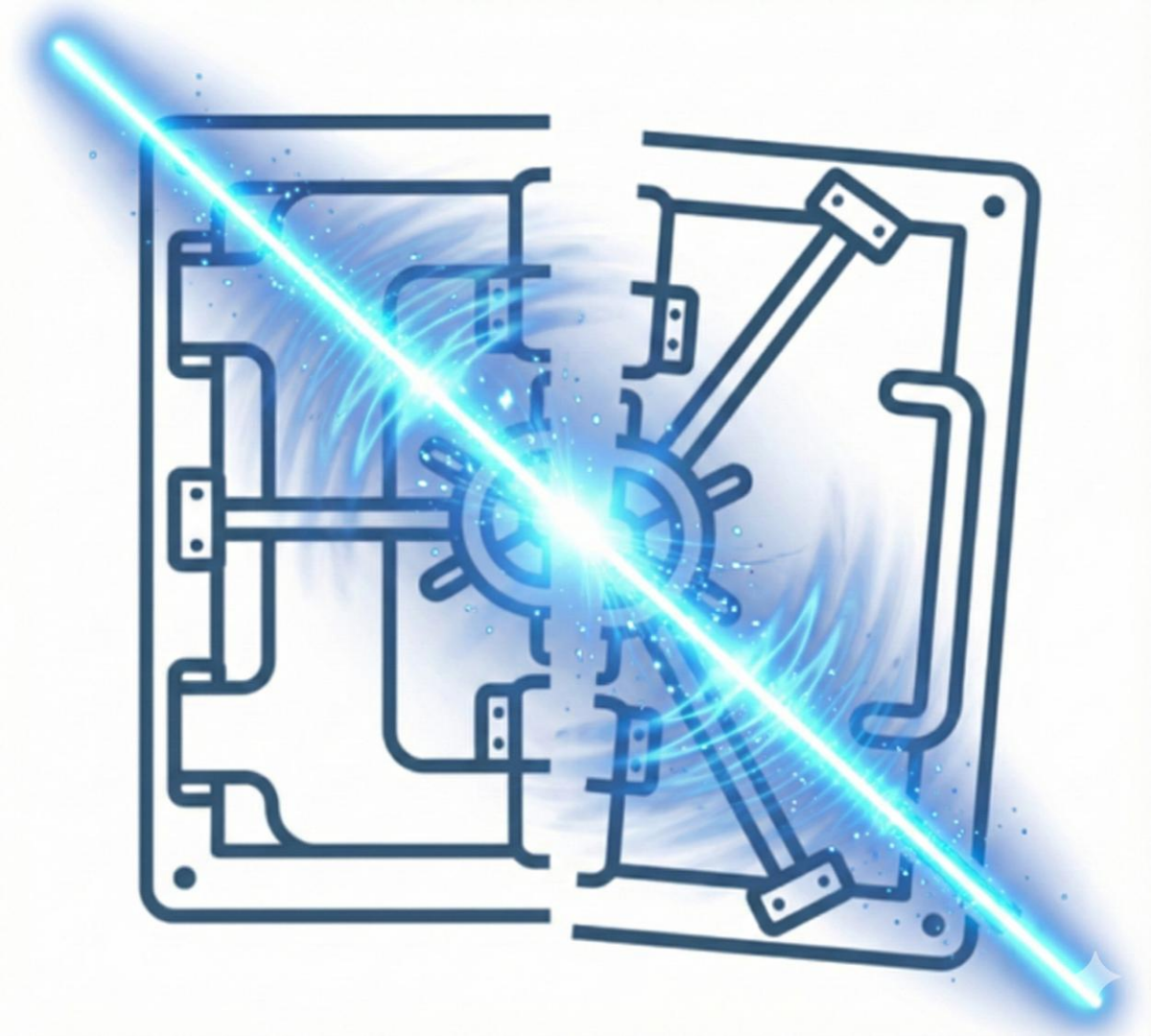
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請注意，儘管我們已盡力羅列各項重要風險，但仍可能存在尚未識別或當前看似不顯著的其他風險因素。

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以數學為基礎的安全性時代已結束

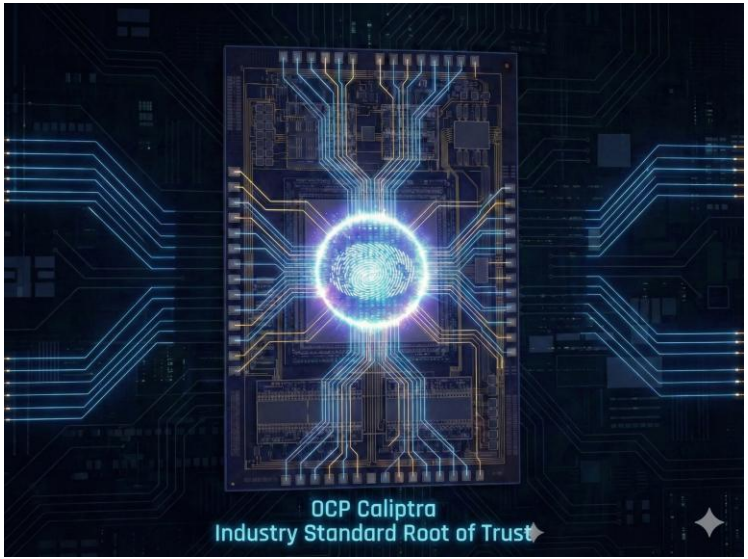
Quantum computers break it in seconds.

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透過 CNSA 2.0 與 Caliptra 達成量子韌性

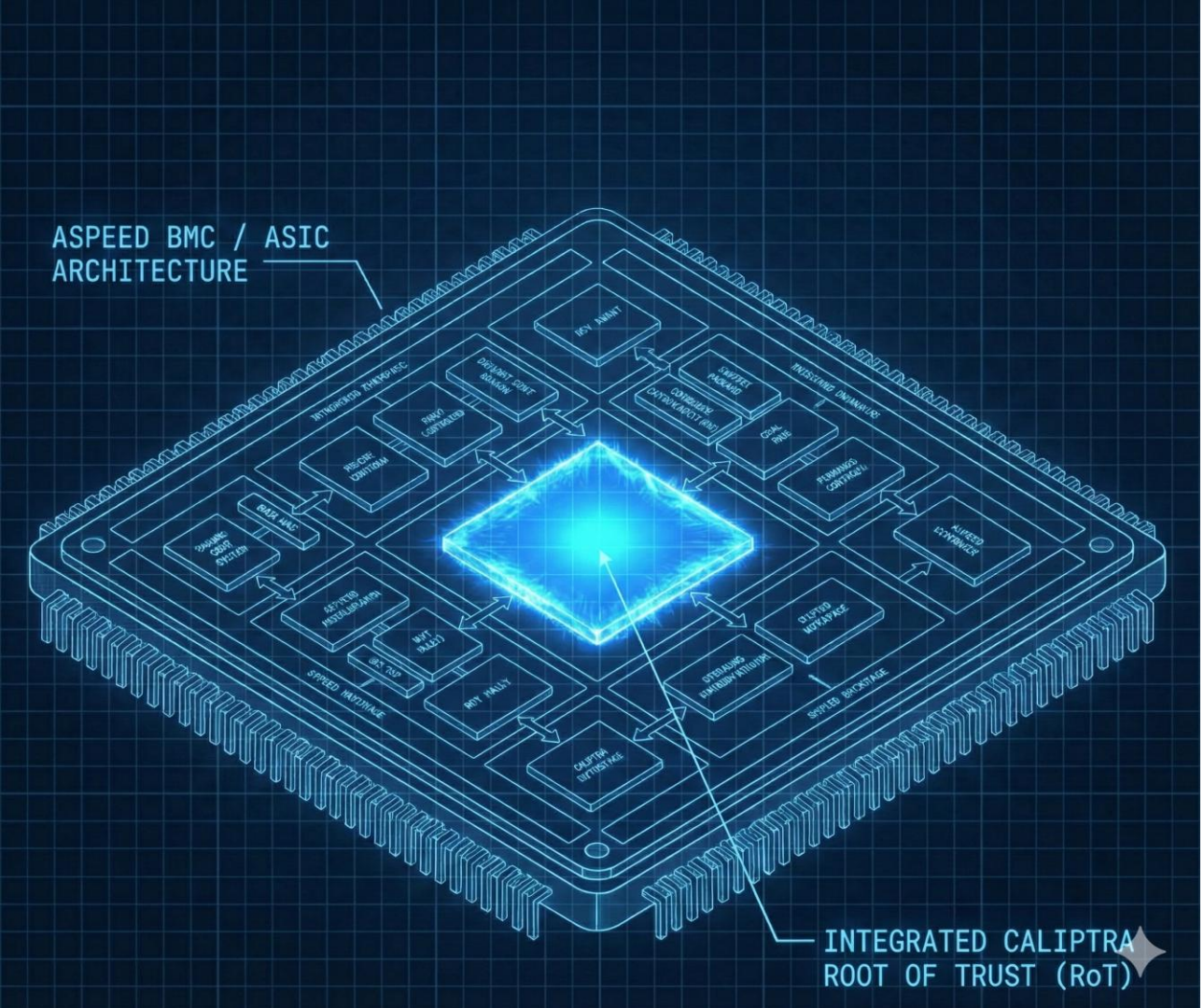


Mandatory cryptographic migration.



Embedding post-quantum security directly into the silicon to create an immutable Root of Trust.

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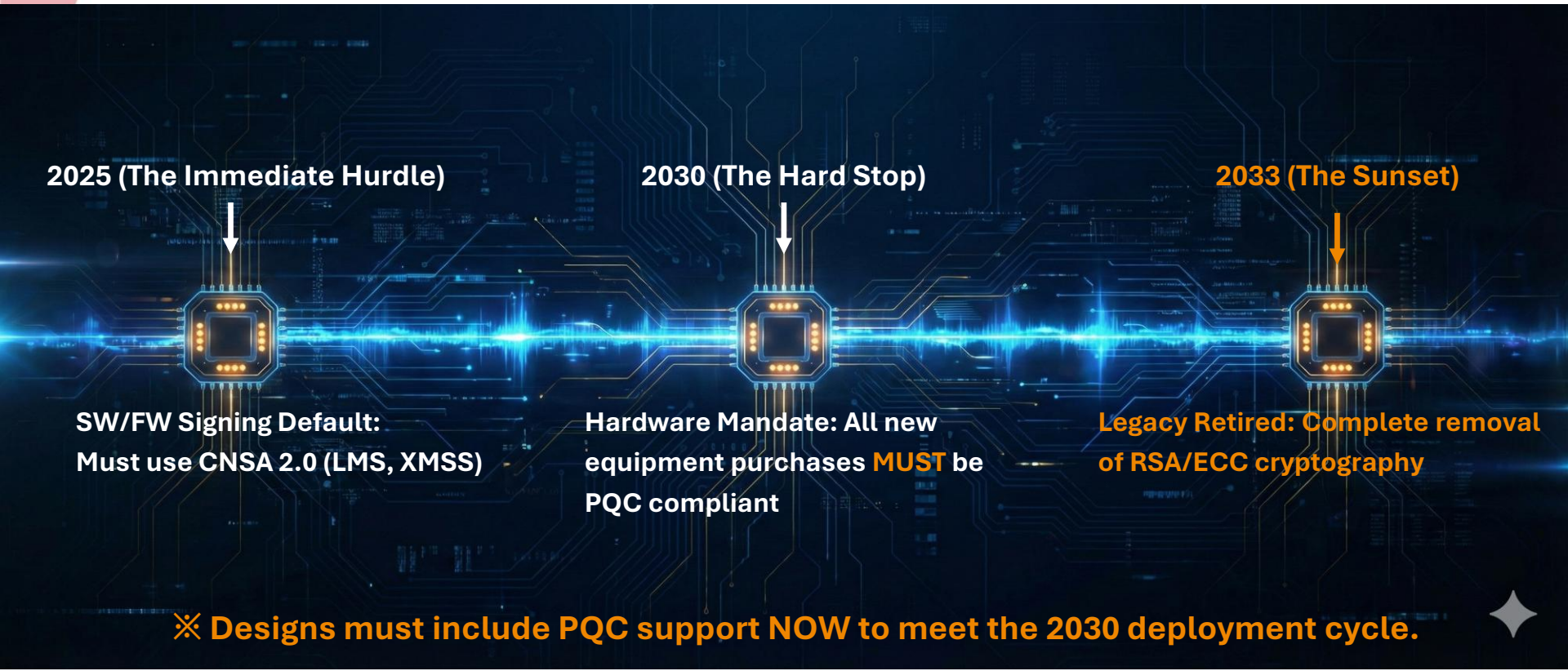
ASPEED BMC 符合 CNSA 2.0

*"**ASPEED** is the first adopter of **OCP Caliptra**, embedding post-quantum security directly into the silicon to create an immutable **Root of Trust**"*

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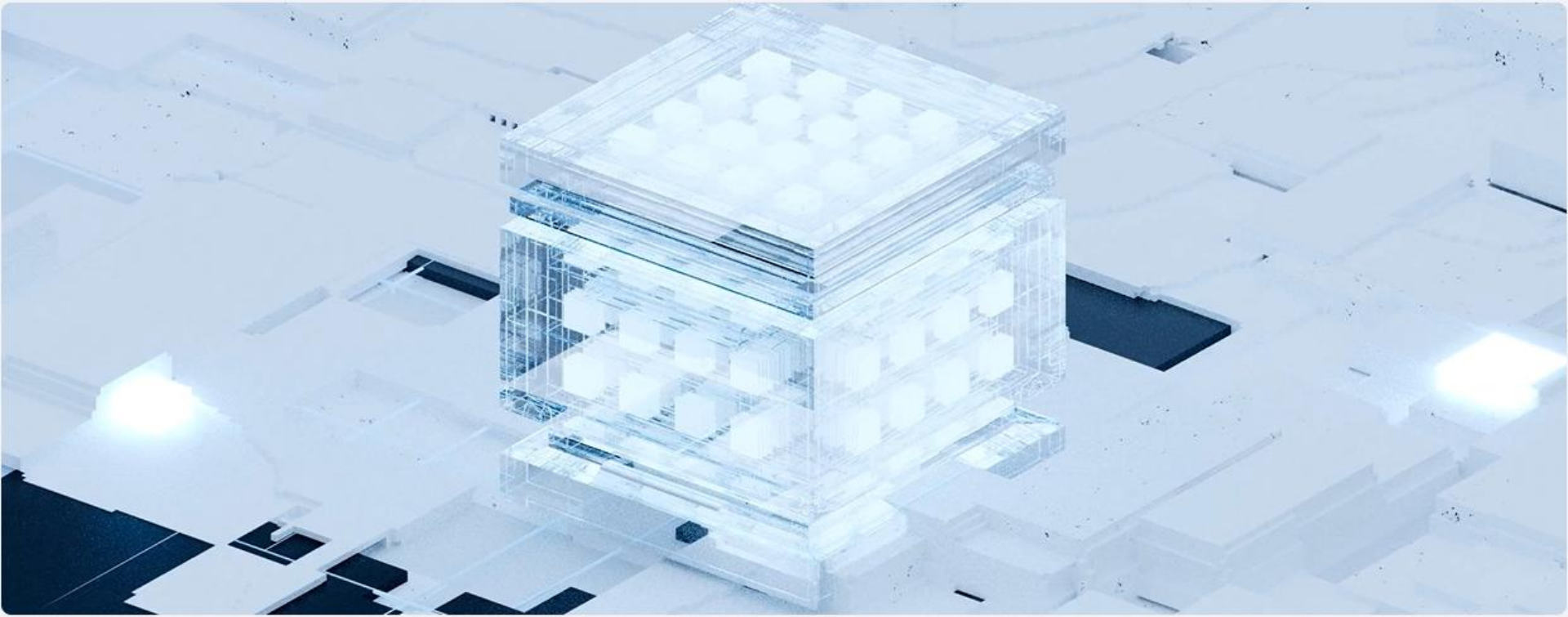
CNSA 時間表

Aggressive. Non-Negotiable. Mandatory.



※ Designs must include PQC support NOW to meet the 2030 deployment cycle.

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Caliptra 由產業領導者支持

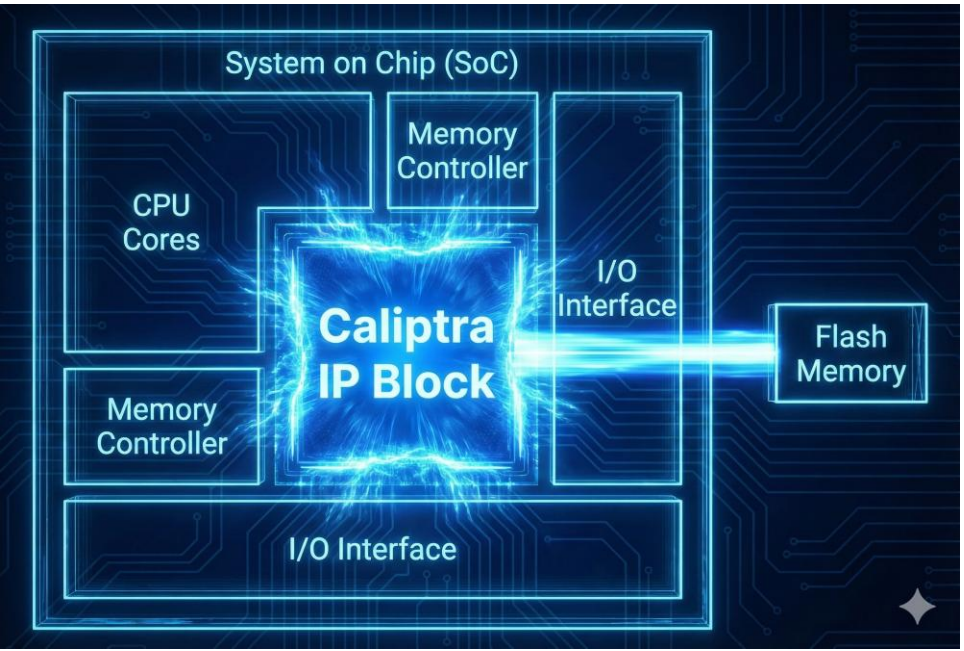
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為 BMC、CPU、GPU、DPU 與 TPU 提供安全基礎

A reusable Hardware IP block embedded directly into the silicon.



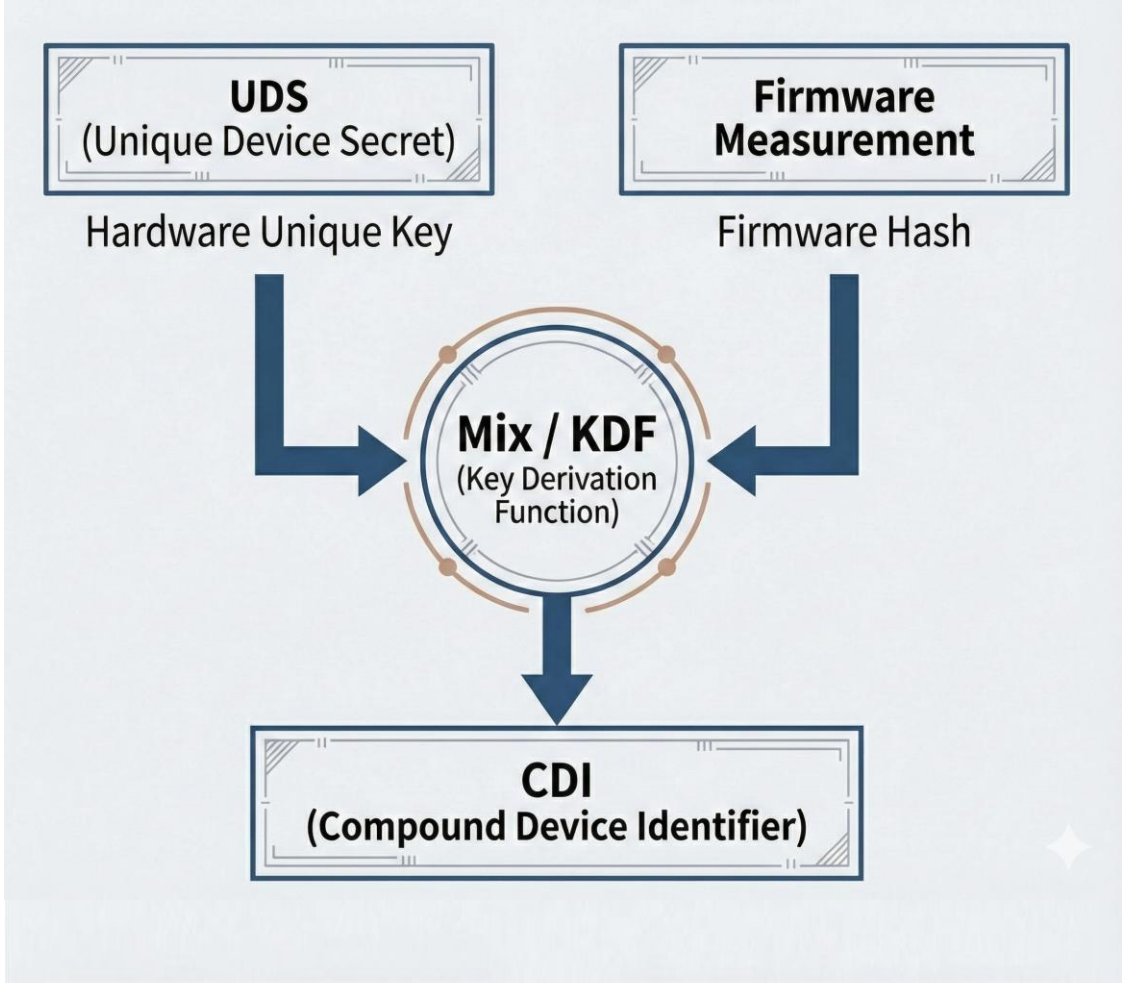
- *Root of Trust for Measurement (RTM)*
- *Secure Boot*
- *PQC algorithms*
- *Caliptra consists of:*
  - ◎ RTL (hardware)
  - ◎ ROM code
  - ◎ Firmware code

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晶片身分來源

DICE – Device Identifier Composition Engine

- *UDS is the hardware unique key, accessible only by Caliptra.*
- *Measurement: hash operations*



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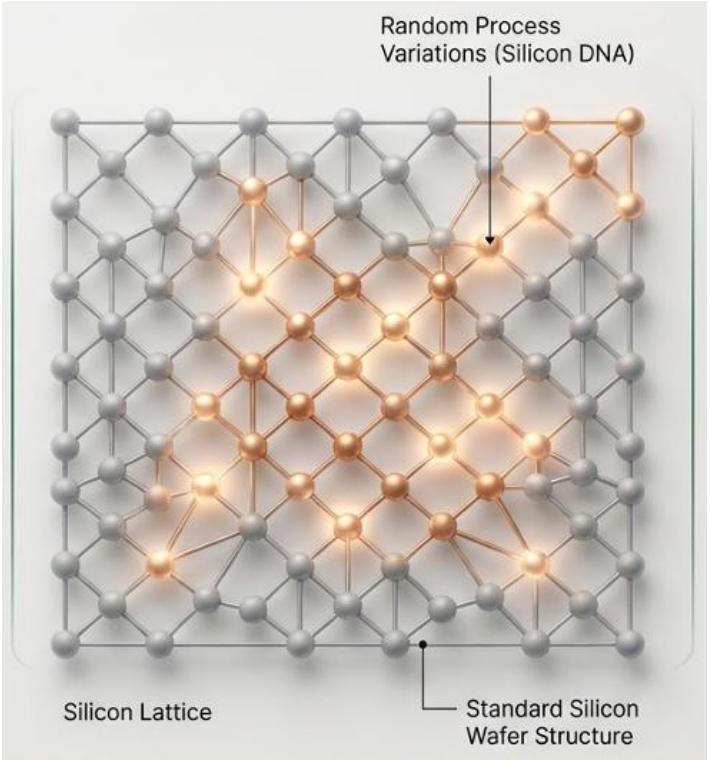
使用 PUF 鞏固 Caliptra 的基礎

Meets Caliptra's requirements for a high-entropy, unique secret.

PUF: Superior Source for the Unique Device Secret (UDS) for Caliptra.

"Even the manufacturer cannot create two identical chips."

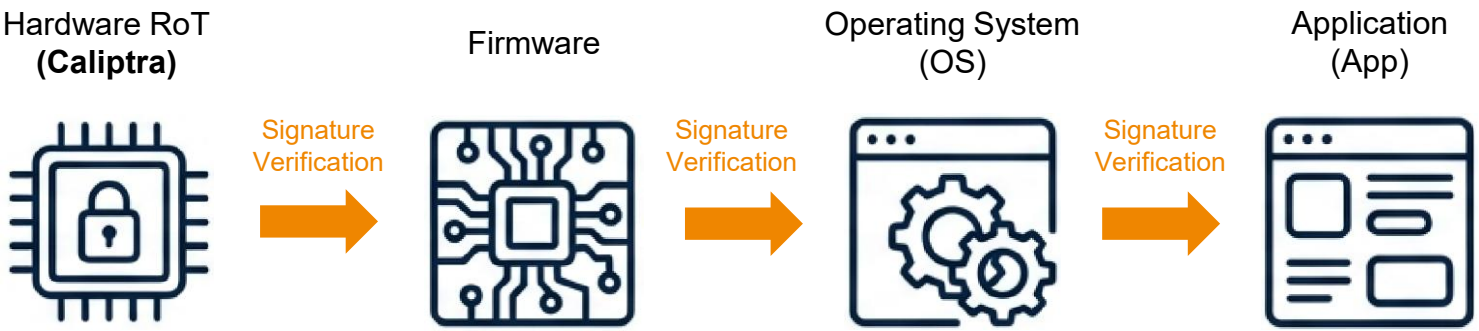
High entropy *powered by PUF is perfect for* ***quantum-resistant*** *architecture.*



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ASPEED 如何建立全面安全完整性 (Total Integrity)

Blocking threats with Chain of Trust.



Secure Boot

Verify first, then execute.

Block low-level malware.

Silicon Root of Trust

Hardware cannot be tampered.

Establish a chain of trust.

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OCP S.A.F.E. 合規性

Standardizing security audits for the supply chain.

OPEN Compute Project SOLUTION PROVIDER®

OCP S.A.F.E.

The Problem: Traditional security audits are fragmented.

The Solution: A standardized framework.

3rd Party Verification: Independent security labs approved by OCP.

Supply Chain Trust: Transparent, objective proof that ASPEED's firmware and hardware meet rigorous security standards.

ASPEED

PQC 持續演進：我們的解決方案具備未來前瞻性

Embracing the dynamic nature of post-quantum cryptography.

The Standard in NOT Static.

Crypto-Agility is Key.

Continuous Monitoring.

Long-Term Commitment.

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融合 PUF、PQC 與 Caliptra 以打造量子抗性資料中心



Caliptra (Standard)

This industry standard Root of Trust is PQC-ready and execute boot measurements.

PUF (Entropy)

True randomness derived from physical silicon variations (the silicon “fingerprint”),

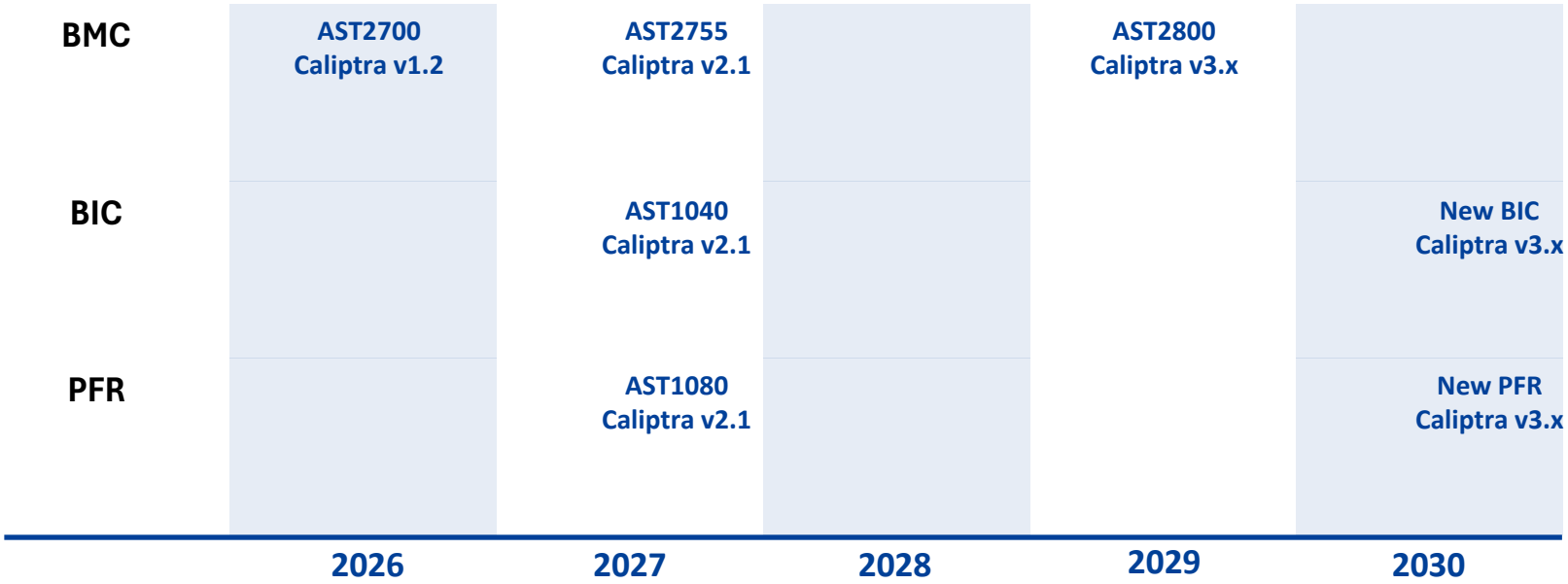
PQC (Algorithm)

NIST-standard algorithms designed to resist quantum attacks, and it requires high-quality entropy.

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量子韌性產品發展路線圖

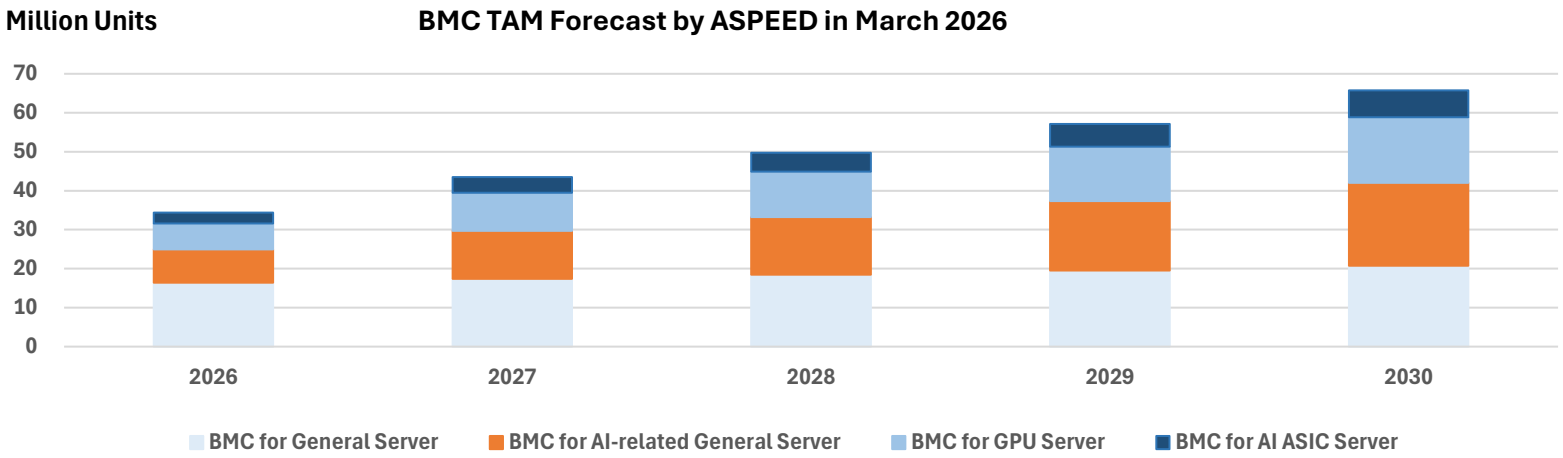
Delivering quantum-resilient ***OCP Caliptra security*** across the entire ***ASPEED*** product portfolio.



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BMC TAM（整體有效市場）預測修正

- We see huge demand for **AI-related General Server** for agentic AI workload and simple inference tasks.
- BMC TAM is estimated to be **65.77 million** units in 2030.



Note：1) BMC for General Servers will grow at 6% CAGR.
2) BMC for AI-related General Servers is estimated to be **8.5 million** in 2026, with 45% growth rate in 2027 and 20% thereafter.
3) BMC for GPU servers and AI ASIC servers in 2026 is based on the estimate of GPU and AI ASIC shipment and the server architecture.
4) Assuming BMC for GPU Servers grow at 45% in 2027 and 20% afterwards. We also include BMCs for ICMS here.
5) Assuming BMC for AI ASIC Servers grow at 45% in 2027 and 20% afterwards.

Source：ASPEED estimate

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2026年第二季營運展望

For 2026 Q2, we currently expect:

- Consolidated Revenue: Around **NT$3.4 billion ~ 3.6 billion**
- Gross margin: **67% ~ 68%**

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